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PolyCrit: An Online Collaborative Platform for Polymer Characterization

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ABSTRACT

Polymer liquid chromatography at critical conditions (LCCC) is a chromatographic separation condition achieved by carefully balancing the interaction of a polymer with stationary and mobile phases to make the elution time of a polymer in chromatography independent of its molecular weight. By removing the dependence of elution time on polymer molecular weight, the LCCC then allows separation of polymer samples on the basis of secondary differences, such as topology, branching, and end-group functionality, that are otherwise difficult to resolve. Despite its potential, LCCC remains under-employed due to the complexity of its optimization and the scattered nature of existing data. To address these challenges, we developed *PolyCrit*, a database that organizes 428 critical chromatography conditions (characterized by 33 parameters) into a searchable and accessible online platform. *PolyCrit* centralizes decades of literature, providing detailed information on polymers, solvents, stationary phases, and chromatographic parameters. It features a quality scoring system to ensure data reliability and supports contributions from the research community through a validation process. By curating experimental critical conditions, *PolyCrit* reduces the need for extensive literature searches to utilize the powerful chromatographic technique. Additionally, *PolyCrit* invites current researchers to contribute to the database by submitting their own work. It can be found at https://lccc.ywangcomp.org.

1. Introduction

Synthetic polymer samples are seldom composed of identical copies of the same molecule; instead, they consist of a distribution of species that vary across multiple characteristics, such as molecular weight (M_w), chemical composition, functionality, chain architecture, and topology. Accurately and comprehensively characterizing these complex polymer systems' distributions is of significant importance to many experimental chemists, industrial researchers, and material scientists. One such umbrella of methodologies for characterization is known as liquid chromatography, and it is a versatile analytical technique essential for separating and analyzing components within a mixture based on their physical or chemical properties [1]. For example, a solution of polymers [2] varying in degrees of polymerization (a property that is proportional to M_w) can be separated by M_w using a method known as Size Exclusion Chromatography (SEC). In SEC, the stationary and mobile phases are chosen to minimize interactions with polymers, preventing their adsorption and ensuring that separation occurs solely based on differences in polymer's hydrodynamic volumes. SEC is commonly used to obtain molecular weight distribution in polymer samples. However, SEC cannot be used to obtain other molecular characteristics' distributions such as functionality, chemical composition, and architecture.

In response to the needs of characterizing these complex polymer systems, analytical polymer chemists have developed Interactive Chromatography (IC) methods that tap into the chemical specific interaction between polymers and column substrate. The collective IC systems include solvent gradient interactive chromatography (SGIC), thermal gradient interactive chromatography (TGIC), Liquid Adsorption Chromatography (LAC), etc. Unlike SEC, where polymer adsorption is minimized, IC systems intentionally allow polymers to adsorb onto the chromatography columns. The adsorption strength is carefully adjusted by modifying the solvent composition or temperature, which in turn enables detailed probing of the polymers' chemical composition and functionality. The fine-tuning of polymer adsorption on column substrates relies on one important concept in polymer physics, the critical adsorption point (CAP), which marks the condition that a given polymer is either adsorbed on the column surface or repelled from the column surface. The existence of CAP leads to a special type of polymer chromatography mode, namely, liquid chromatography at the critical condition (LCCC). At the LCCC condition, a polymer will elute independent of its molecular weight. In plots of elution time versus M_w, this behavior is distinctly represented by a vertical line (Fig. 1). At this point, the species' interaction with the stationary phase balances its interaction with the mobile phase (refer to Section 2 for more information). This balance enables separation based on subtle differences in

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